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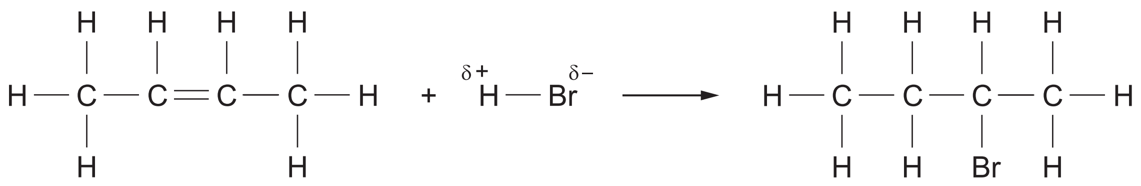
SECTION B

Answer **all** questions.

6. Butanone can be prepared from but-2-ene using a three-step synthesis.

(a) In the first step, but-2-ene is reacted with HBr to form 2-bromobutane.



(i) Circle the species that represents the electrophile. [1]

(ii) Name the type of bond fission that takes place in the H — Br bond in the first step of the mechanism. [1]

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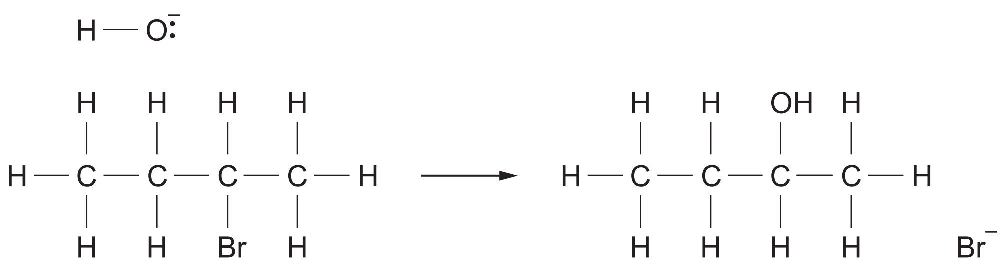
(b) In the second step, 2-bromobutane undergoes nucleophilic substitution to form butan-2-ol.



(i) Use curly arrows to complete the equation to show the mechanism of the nucleophilic substitution. Include any relevant partial charges. [2]

(ii) Give the reagents and conditions required for this nucleophilic substitution. [2]

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(iii) State the classification of alcohol to which butan-2-ol belongs. [1]

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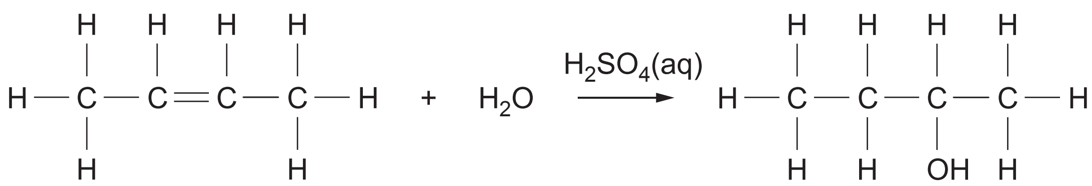
(c) In the final step, butan-2-ol is heated with acidified potassium manganate(VII) to produce butanone.

(i) State the role of the acidified potassium manganate(VII) in this reaction. [1]

(ii) Explain why butanone can be removed from the reaction as it is formed using distillation, leaving unreacted butan-2-ol in the reaction mixture. [2]

(d) Butan-2-ol can also be made directly by hydration of but-2-ene in the presence of dilute sulfuric acid, which acts as a catalyst.



(i) Suggest why the overall yield of the two-step synthesis is likely to be lower than the yield of the direct hydration. [1]

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(ii) Reaction of butan-2-ol with **concentrated** sulfuric acid results in the formation of three isomers with the formula C_4H_8.

I. Name the type of reaction used to form alkenes from alcohols. [1]

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II. Give the structure and **name** of the three isomers formed. [3]

Structure	Structure	Structure
Name:	Name:	Name: ....

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